

Hassan Mazin Makki -A- HW 6

Given: $S = 5000 \text{ kVA}$, $V_L = 10,000 \text{ V}$, $N_s = 1500 \text{ rpm}$, $f = 50 \text{ Hz}$, $X_s = 20\%$, $\delta_m = 0.5^\circ$

Find: Synchronizing power per mech. angle and synchronizing torque at $\delta_m = 0.5^\circ$ for (a) no-load (b) full-load pf = 0.8 lag

Preliminary Calculations

$$V_\phi = \frac{V_L}{\sqrt{3}} = \frac{10,000}{\sqrt{3}} = 5773.5 \text{ V} \quad p = \frac{120f}{N_s} = \frac{120 \times 50}{1500} = 4 \text{ poles}$$

$$I_{FL} = \frac{S}{\sqrt{3} V_L} = \frac{5 \times 10^6}{\sqrt{3} \times 10^4} = 288.7 \text{ A} \quad Z_{base} = \frac{V_\phi}{I_{FL}} = 20 \Omega \Rightarrow X_s = 0.20 \times 20 = 4 \Omega$$

$$\omega_s = \frac{2\pi N_s}{60} = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/s}$$

Electrical displacement for $\delta_m = 0.5^\circ$:

$$\alpha = \frac{p}{2} \times \delta_m = 2 \times \left(0.5 \times \frac{\pi}{180}\right) = 0.01745 \text{ elec. rad}$$

Key Formulas (Sec. 1.10.2)

For two identical parallel machines ($R_a = 0$, small angle α):

$$I_{sy} = \frac{E_r}{X_{st}} = \frac{\alpha E_f}{X_{st}} \quad P_{sy}^{(\text{per phase, 1 machine})} = E_f I_{sy} = \frac{\alpha E_f^2}{X_{st}}$$

$$3P_{sy}^{(\text{total, 2 machines})} = 2 \times 3 \times \frac{\alpha E_f V_\phi}{X_s} \quad T_{sy} = \frac{3P_{sy}}{\omega_s}$$

(At no-load: $E_f = V_\phi \Rightarrow E_f V_\phi = E_f^2$)

Part (a) — No-Load

At no-load: $E_f = V_\phi = 5773.5 \text{ V}$

$$3P_{sy} = \frac{2 \times 3 \times \alpha \times E_f \times V_\phi}{X_s} = \frac{2 \times 3 \times 0.01745 \times 5773.5 \times 5773.5}{4}$$

$$= \frac{2 \times 3 \times 0.01745 \times 33,333,333}{4} = \frac{3,490,000}{4} \times 6 = \frac{3 \times 1,745,000 \times 2}{1} = 872,663 \text{ W}$$

$$3P_{sy} \text{ (no-load)} \approx 872.8 \text{ kW}$$

$$T_{sy}^{(\text{total})} = \frac{3P_{sy}}{\omega_s} = \frac{872,663}{157.08}$$

$$T_{sy}^{(\text{total})} \approx 5556.4 \text{ N} \cdot \text{m} \quad T_{sy}^{(\text{per machine})} = \frac{5556.4}{2} \approx 2778.2 \text{ N} \cdot \text{m}$$

Part (b) — Full-Load, pf = 0.8 Lagging

Step 1: Find E_f and torque angle δ

$$\cos \phi = 0.8, \quad \sin \phi = 0.6, \quad R_a = 0 \text{ (neglected)}$$

$$\begin{aligned} E_f &= \sqrt{(V_\phi \cos \phi)^2 + (V_\phi \sin \phi + I_{FL} X_s)^2} \\ &= \sqrt{(5773.5 \times 0.8)^2 + (5773.5 \times 0.6 + 288.7 \times 4)^2} = \sqrt{(4618.8)^2 + (3464.1 + 1154.8)^2} \\ &= \sqrt{(4618.8)^2 + (4618.9)^2} = \sqrt{42,667,702} \end{aligned}$$

$$E_f = 6532 \text{ V}$$

$$\tan \delta = \frac{V_\phi \sin \phi + I X_s}{V_\phi \cos \phi} = \frac{4618.9}{4618.8} = 1.000 \quad \Rightarrow \quad \delta = 45^\circ$$

Step 2: Synchronizing Power

$$\begin{aligned} 3P_{sy} &= \frac{2 \times 3 \times \alpha \times E_f \times V_\phi}{X_s} = \frac{2 \times 3 \times 0.01745 \times 6532 \times 5773.5}{4} \\ &= \frac{6 \times 0.01745 \times 37,720,302}{4} = \frac{3,948,973}{4} = 987,243 \text{ W} \end{aligned}$$

$$3P_{sy} \text{ (full-load)} \approx 987 \text{ kW}$$

Step 3: Synchronizing Torque

$$T_{sy}^{(\text{total})} = \frac{987,243}{157.08}$$

$$T_{sy}^{(\text{total})} \approx 6283 \text{ N} \cdot \text{m} \quad T_{sy}^{(\text{per machine})} = \frac{6283}{2} \approx 3141.7 \text{ N} \cdot \text{m}$$

Results Summary

Quantity	Formula	(a) No-Load	(b) Full-Load 0.8 lag
E_f	phasor	5773.5 V	6532 V
δ	phasor	0°	45°
$3P_{sy} \text{ total}$	$\frac{2 \cdot 3 \cdot \alpha \cdot E_f V_\phi}{X_s}$	872.8 kW	987 kW
$T_{sy} \text{ total}$	$3P_{sy}/\omega_s$	5556.4 N·m	6283 N·m
$T_{sy} \text{ per machine}$	$T_{sy}/2$	2778.2 N·m	3141.7 N·m